

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DEEKSHITHA U
Register Number	192424055
Date	03 October 2026
Day	Saturday

SET 9 — Online Learning Activity Analysis

Aim

To analyze online learning activity using **R programming and Tableau** by studying quiz performance, study time, videos watched, login dates, and course-wise performance.

Dataset

Student ID	Gender	Age	Course	Study Time (hrs)	Videos Watched	Quiz Score	Login Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

Q1. Histogram and Boxplot of Quiz Scores

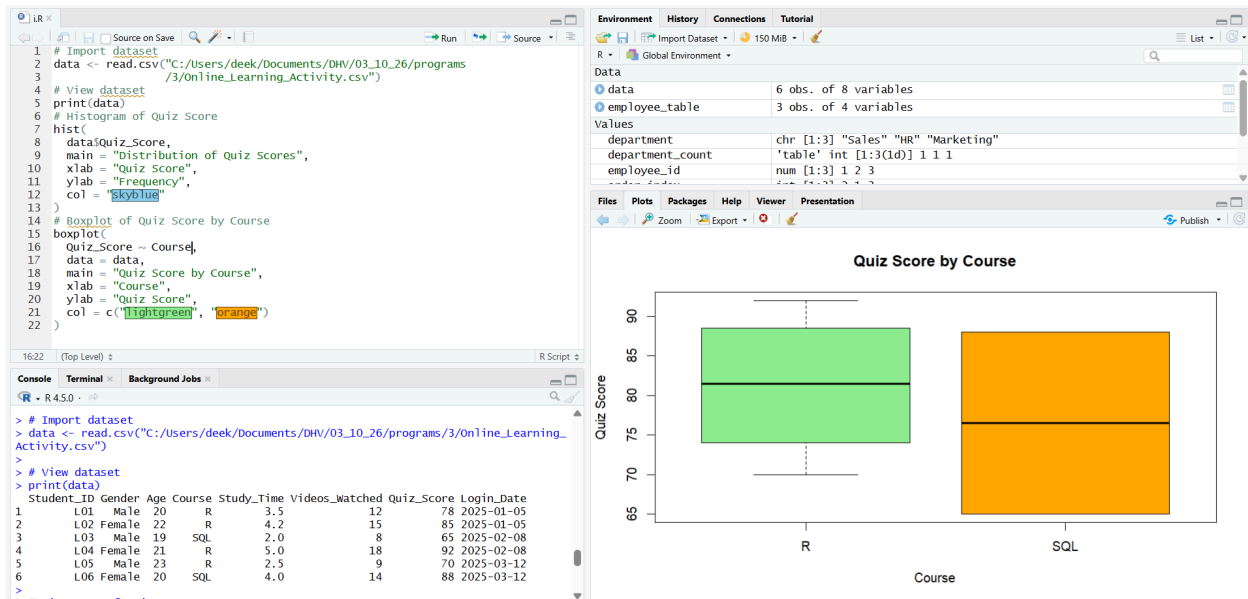
Question

Import the dataset in R. Plot a histogram for **Quiz_Score**. Draw a boxplot of **Quiz_Score** by **Course**. Add title and labels. Interpret student performance.

R Program

```
# Import dataset
data <- read.csv("Online_Learning_Activity.csv")
# View dataset
print(data)
# Histogram of Quiz Score
hist(
  data$Quiz_Score,
  main = "Distribution of Quiz Scores",
  xlab = "Quiz Score",
  ylab = "Frequency",
  col = "skyblue"
)
# Boxplot of Quiz Score by Course
boxplot(
  Quiz_Score ~ Course,
  data = data,
  main = "Quiz Score by Course",
  xlab = "Course",
  ylab = "Quiz Score",
  col = c("lightgreen", "orange")
)
```

Output



Observation

- Quiz scores range from **65 to 92**.
- The highest score is **92**.
- The lowest score is **65**.
- The R-course scores are 78, 85, 92, and 70.
- The SQL-course scores are 65 and 88.
- There is variation in performance within both courses.

Q2. Scatter Plot — Study Time vs Quiz Score

Question

Create a scatter plot using R programming between **Study_Time** and **Quiz_Score**. Represent **Videos_Watched** using bubble size. Apply transparency. Explain the learning pattern.

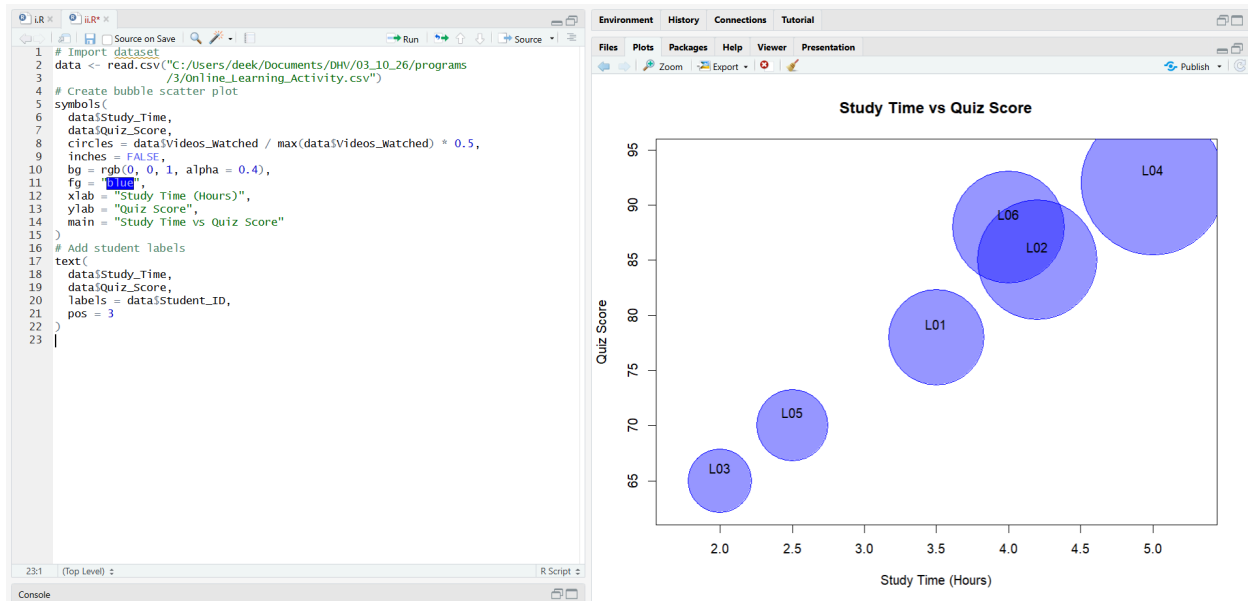
R Program

```
# Import dataset
data <- read.csv("Online_Learning_Activity.csv")

# Create bubble scatter plot
symbols(
  dataStudy_time, dataQuiz_Score,
  circles = dataVideos_watched/max(dataVideos_Watched) * 0.5,
  inches = FALSE,
  bg = rgb(0, 0, 1, alpha = 0.4),
  fg = "blue",
  xlab = "Study Time (Hours)",
  ylab = "Quiz Score",
  main = "Study Time vs Quiz Score"
)

# Add student labels
text(
  dataStudy_time, dataQuiz_Score,
  labels = data$Student_ID,
  pos = 3
)
```

Output



Observation

The data generally suggests that students with greater study time and more videos watched can have higher quiz scores. For example, the student studying **5 hours** and watching **18 videos** scored **92**.

However, the relationship is not perfectly consistent. For example, one student studying 4 hours scored 88, while another studying 4.2 hours scored 85. Therefore, study time and videos watched appear to be associated with performance in this small dataset, but they do not alone determine the quiz score.

Q3. Monthly Average Quiz Score and Moving Average

Question

Using R programming, convert **Login_Date** into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

R Program

```
# Import dataset
data <- read.csv("Online_Learning_Activity.csv")

# Convert Login_Date into Date format
dataLogin_date <- as.Date(dataLogin_Date)

# Extract month
dataMonth <- format(dataLogin_Date, "%Y-%m")

# Calculate average quiz score per month
monthly_avg <- aggregate(
  Quiz_Score ~ Month,
  data = data,
  FUN = mean
)

print(monthly_avg)

# Plot monthly average quiz score
plot(
  1:nrow(monthly_avg),
  monthly_avg
  Quiz_score, type = "o", pch = 19, xaxt = "n", xlab = "Month", ylab = "AverageQuizScore"
  Month
)

# 2-month moving average
moving_avg <- stats::filter(
  monthly_avg$Quiz_Score,
  rep(1/2, 2),
  sides = 1
)

print(moving_avg)

# Add moving average to graph
```

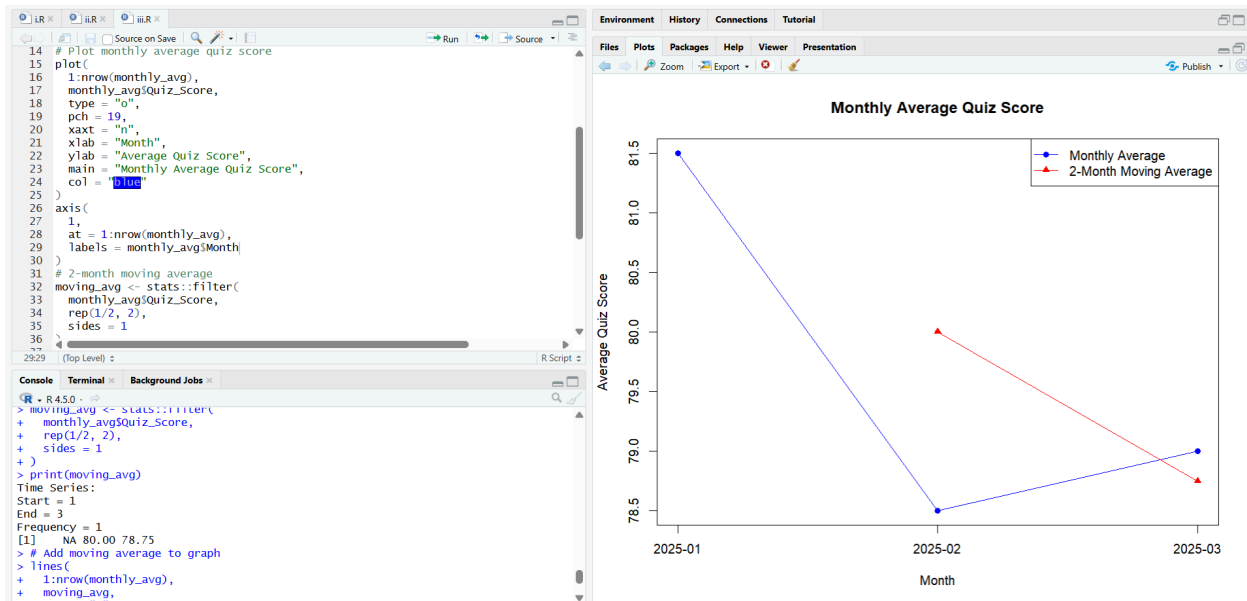
```

lines(
  1:nrow(monthly_avg),
  moving_avg,
  type = "o",
  pch = 17,
  col = "red"
)

legend(
  "topright",
  legend = c("Monthly Average", "2-Month Moving Average"),
  col = c("blue", "red"),
  lty = 1,
  pch = c(19, 17)
)

```

Output



Observation

- Average quiz score decreased from **81.50 in January** to **78.50 in February**.
- It increased slightly to **79.00 in March**.
- Overall, the performance shows a **small decline followed by a slight recovery**.
- The moving average provides a smoother view of the monthly performance pattern.

Q4. Tableau Interactive Dashboard

Question

Import *Online_Learning_Activity.csv* into Tableau. Create:

- Bar chart showing average quiz score by Course.
- Scatter plot of Study Time vs Quiz Score.
- Filter for Gender and Course.
- Combine them into an interactive dashboard.

Tableau Procedure

Step 1 — Import Dataset

1. Open **Tableau**.
2. Select **Text File**.
3. Select *Online_Learning_Activity.csv*.
4. Check that the fields are recognized correctly.

Step 2 — Create Bar Chart

Create a new worksheet.

- Drag **Course** → Columns.
- Drag **Quiz_Score** → Rows.
- Change aggregation to **Average**.
- Select **Bar** from the Marks card.
- Add a suitable title:

Average Quiz Score by Course

Step 3 — Create Scatter Plot

Create another worksheet.

- Drag **Study_Time** → Columns.
- Drag **Quiz_Score** → Rows.
- Select **Circle** from Marks.
- Drag **Videos_Watched** → Size.
- Drag **Gender** → Color.

Title:

Study Time vs Quiz Score

Step 4 — Add Filters

Add:

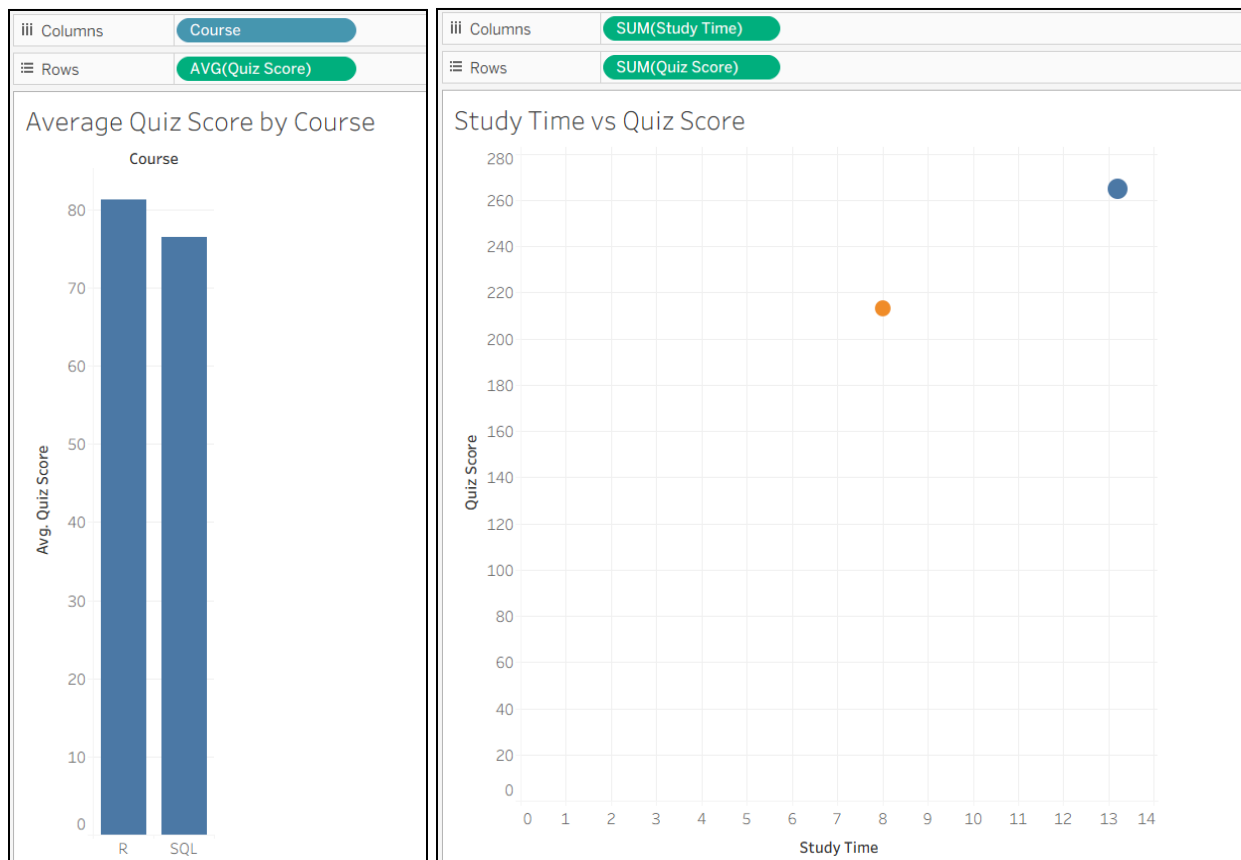
- **Gender** filter
- **Course** filter

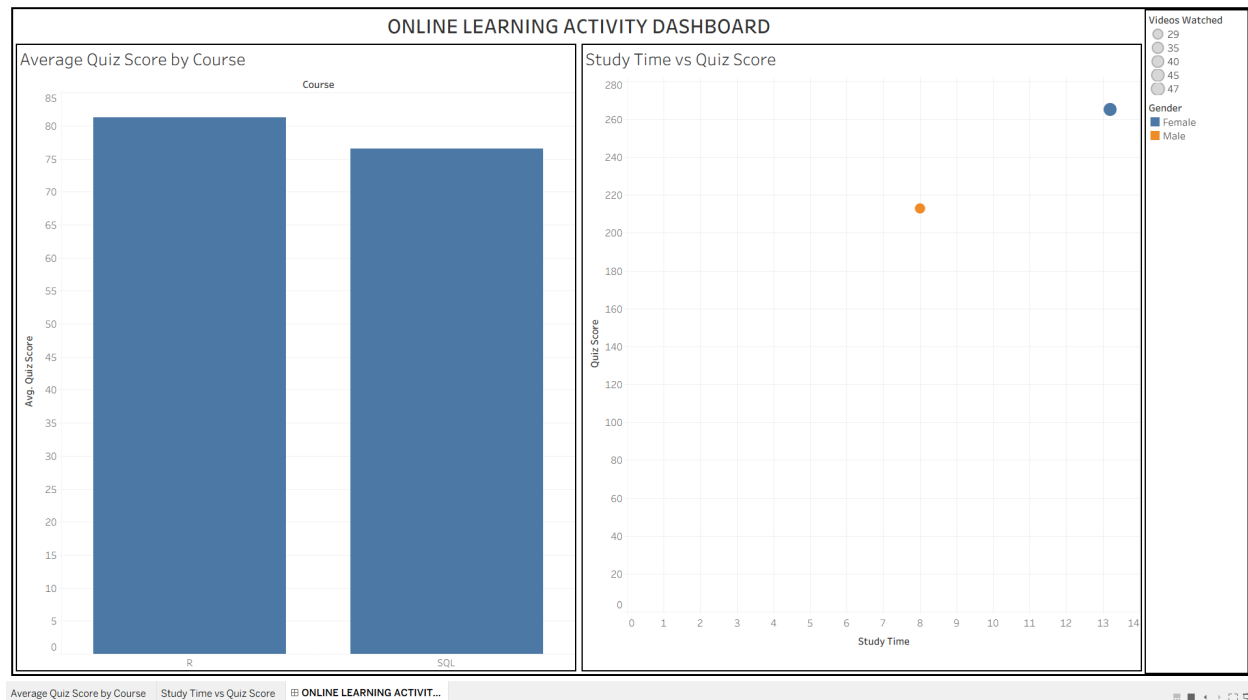
These filters allow the user to explore the data for specific genders or courses.

Step 5 — Create Dashboard

1. Select **New Dashboard**.
2. Add the **Average Quiz Score by Course** bar chart.
3. Add the **Study Time vs Quiz Score** scatter plot.
4. Add the **Gender** and **Course** filters.
5. Arrange the charts neatly.
6. Add the title:

Online Learning Activity Dashboard





Procedure

1. Open **RStudio** and import **Online_Learning_Activity.csv** using **read.csv()**.
2. Examine the dataset and identify the variables required for analysis.
3. Use **hist()** to create a **histogram of Quiz Score**.
4. Use **boxplot()** to compare **Quiz Score by Course**.
5. Use **symbols()** to create a **bubble scatter plot** between Study Time and Quiz Score, using Videos Watched to control bubble size.
6. Convert **Login_Date** into R's Date format using **as.Date()**.
7. Extract the month and calculate the **average quiz score for each month** using **aggregate()**.
8. Create a line chart of monthly average quiz scores and add a moving-average smoothing line.
9. Open **Tableau** and import **Online_Learning_Activity.csv**.
10. Create a bar chart for **average quiz score by course**.
11. Create a scatter plot for **Study Time vs Quiz Score** and represent Videos Watched using bubble size.
12. Add **Gender** and **Course** filters.
13. Combine the visualizations into an interactive Tableau dashboard.

Result

The **Online Learning Activity** dataset was successfully analyzed using R and Tableau. A histogram and boxplot were created to study quiz-score performance, a bubble scatter plot was created to analyze the relationship between study time and quiz score, and monthly average quiz scores with moving-average smoothing were visualized. Finally, an **interactive Tableau dashboard** containing course-wise average quiz scores, a study-time versus quiz-score scatter plot, and Gender/Course filters was successfully developed.